Pranav Gupta

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BEL Intern – SAR Processing | State-level Roll Ball captain | JEE Mains 92.3 Percentile

Degree	Board	Institute	% / CGPA	Year
B.Tech (CSE)	Shiv Nadar University	Shiv Nadar Institute of Eminence	**7.62/10**	2028
Class XII	CBSE	Seth Anandram Jaipuria School, Ghaziabad	**93%**	2024
Class X	CBSE	Amity Internation School Sector-6, Vasundhara	**92%**	2022

ACADEMIC ACHIEVEMENTS

ASME - SNIoE	• Selected among teams nationwide to compete in ASME SDC 2024-25, an all-India engineering design challenge attracting 100+ teams across premier institutions and **securing 7th position**	2025
JEE Mains	• Ranked in **top 7.7% of 1.2 million+** JEE Mains 2024 aspirants | Secured 93.2% in Class XII and 94.2% in Class X	2024
Duke of Edinburgh – Bronze Award	• Completed Duke of Edinburgh's International Bronze Award (2022), recognizing achievement across volunteering, skill development, and expedition — awarded to select candidates globally	2021

INTERNSHIPS

Central Research Laboratory (BEL) – GAD — **Jun'25-Aug'25**

Intern	**SAR Processing**	• Engineered **Python-based** data processing pipelines for **Sentinel-1 SAR** satellite datasets, applying signal processing and large-scale structured data analysis techniques directly transferable to ML data pipelines • Implemented automated data visualization workflows in Python and C, developing strong foundations in data preprocessing, feature extraction, and handling high-dimensional real-world datasets • Built geospatial navigation applications using **Copernicus satellite imagery**, applying algorithmic optimization and spatial data reasoning to route planning solutions across India • Gained deep exposure to sensor data acquisition, noise filtering, and signal-to-feature transformation — core preprocessing skills applicable to **ML model training pipelines**
	UAV & eVTOLs	• Gained hands-on familiarity with **QGroundControl (QGC) GCS and MAVLink communication protocol** for UAV telemetry, data exchange, and real-time monitoring between drone and ground systems • Explored UAV autonomy stack basics including flight controller (Pixhawk) configuration and drone-GCS data pipelines — **foundational knowledge for autonomous systems and AI-driven UAV navigation research**

DRIIV: Delhi Research Implementation and Innovation – IIT Delhi — **Dec'24-Jan'25**

Research Intern	• Conducted in-depth research on UAV, UAS, eVTOL, and UTM systems, authoring a **200-page report** synthesizing technological advancements, regulatory frameworks, and integration challenges across India's aerial vehicle ecosystem • Investigated innovations in embedded software, electronics, and materials driving UAV autonomy and performance, while evaluating **government schemes and MSME frameworks** through structured data analysis and strategic reasoning • Developed and released a technical **blog** (www.driiv.co.in/MS/Indowings/) translating complex unmanned avionics research into accessible content, demonstrating technical communication skills valued in AI research and documentation roles • Gained expertise in technical research, strategic analysis, and data visualization, directly contributing to DRIIV's mission of bridging **cutting-edge research and real-world implementation**

PROJECTS

Circuit Designer AI *(Natural Language to Circuit Generator)*	• Built an LLM application using APIs to convert **natural language into structured circuit representations**, enforcing deterministic outputs via strict JSON schemas for production AI systems • Implemented **metadata-driven RAG pipeline** grounding LLM outputs in electronics reference data to reduce hallucinations, with modular prompt orchestration, schema validation, and error handling in a Node.js backend	Ongoing
InboxWhisper *(LangGraph-based Academic Email Task Manager)*	• Designed a LangGraph agentic pipeline ingesting academic emails via Microsoft Graph API, extracting structured entities — deadlines, course codes, locations — using GPT-4o-mini with heuristic fallbacks for reliability • Built a RAG-style chatbot over a SQLite task database with LangSmith tracing for prompt debugging and pipeline observability, **delivered as a full Streamlit application** with priority scoring and calendar export	2025
StreetGPT *(Real-time Generative Scene Captioning)*	• Architected a real-time multimodal AI pipeline integrating OpenAI Vision API for live object detection, scene captioning, and mood analysis from webcam feed, with dynamic prompt generation overlaid on live video frames • Designed a modular LLM-powered system with distinct stages for capture, intelligent region sampling, vision analysis, and prompt orchestration — applying **production-grade AI pipeline** design principles	2025
ASME SDC *(Autonomous Nut Sorter)*	• Represented Shiv Nadar University at ASME SDC 2024-25, an all-India engineering design competition; engineered a fully autonomous ball-bearing sorting system with multi-sensor material and size classification • Designed end-to-end autonomous classification and actuation pipeline delivering sorted bearings to 6 designated stations — hands-on experience in **autonomous systems directly relevant to AI-driven robotics**	2024

EXTRA CURRICULAR ACHIEVEMENTS

Roll Ball Skating	• Served as State-level Captain of skating roll-ball team | Led team in regional and national tournaments	2019-22
Basketball	• Member of school basketball team at interschool competitions across two schools over 3 years	2021-24
Volunteering	• Volunteer at Girl Up Shiv Nadar University (Human Rights) | Duke of Edinburgh's Award (Poverty Alleviation)	Present

TECHNICAL SKILLS

AI & LLMS	Prompt Engineering | JSON Schemas | LLM Evaluation & Response Validation | Constrained Generation | OpenAI API
RAG Frameworks	LangGraph | LangChain | Retrieval-Augmented Generation (RAG) | LangSmith | Microsoft Graph API
Computer Vision	OpenCV | MediaPipe | Real-time Gesture Recognition | Landmark Detection
Tools	Git | GitHub | Jupyter Notebooks | Streamlit | SQLite | Linux | CAD | Vibe-coding
Languages	Python | JavaScript (Node.js) | C | SQL | TypeScript | HTML/CSS

BADGES

Data Science Orientation (2026); **AWS Cloud Foundation** (2026); **AWS Cloud Architecting (**2026)